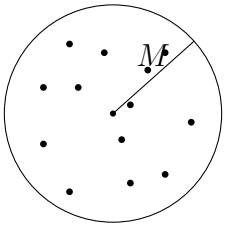
1 Bounded sequences

Definition 1.1. A sequence (a_n) of complex numbers is **bounded** if

$$\exists M > 0 \text{ such that } |a_n| \leq M \ \forall n \in \mathbb{N} \text{ (equivalently } a_n \in D(0, M) \ \forall n \in \mathbb{N}). \tag{1}$$

Remark 1.1. Statement (1) is equivalent to the set $A = \{|a_n|, n \in \mathbb{N}\}$ of the moduli of the terms of the sequence (a_n) being upper bounded.

Remark 1.2. In $\mathbb{R}$, (1) says that the terms of the sequence (a_n) belong to the interval $[-M, M]$ for all $n \in \mathbb{N}$.
In $\mathbb{C}$, (1) says that the terms of the sequence (a_n) belong to the closed ball $D(0, M) = \{z \in \mathbb{C} : |z| \leq M\}$ for all $n \in \mathbb{N}$. Since $D(0, M) \subseteq B(0, M + 1)$, we could equally well ask for the terms to lie in some open ball around 0.



Theorem 1.1. (a_n) converges $\Rightarrow$ (a_n) is bounded.

Proof. Suppose $a_n \to a$ with $a \in \mathbb{C}$. Let $\varepsilon = 1$, there exists $n_0 \in \mathbb{N}$ such that $|a_n - a| < 1 \ \forall n \geq n_0$ (*). Let

$$M = \max\{|a_1|, |a_2|, \ldots, |a_{n_0}|, 1 + |a|\}.$$

Then

$$|a_n| \leq M \ \forall n \leq n_0 \qquad \text{by definition of M,}$$
$$|a_n| = |(a_n - a) + a| \leq |a_n - a| + |a| < 1 + |a| \leq M \ \forall n \geq n_0 \qquad \text{using the triangle inequality, (*), and the definition of M}$$

Thus, $|a_n| \leq M \ \forall n \in \mathbb{N}$ and we conclude that (a_n) is bounded. □

Remark 1.3. Note that the reverse implication is not true, for example $(-1)^n$ is bounded but does not converge.

2 Comparison and Squeeze theorem

Theorem 2.1 (Comparison Theorem). Let (a_n) and (b_n) be **real** sequences such that $a_n \to a$ and $b_n \to b$. If $\exists N \in \mathbb{N}$ such that $a_n \leq b_n \ \forall n \geq N$, then $a \leq b$.

Proof. Suppose, for the sake of contradiction, that $a > b$. Define $\varepsilon = \frac{a-b}{2} > 0$. Since $a_n \to a$, there exists $n_1 \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for all $n \geq n_1$; since $b_n \to b$, there exists $n_2 \in \mathbb{N}$ such that $|b_n - b| < \varepsilon$ for all $n \geq n_2$. Let $n_0 = \max\{N, n_1, n_2\}$. Then for all $n \geq n_0$ we have

$$|a_n - a| < \varepsilon \ \text{ and } \ |b_n - b| < \varepsilon.$$

In particular, this implies that $a_n - a > -\varepsilon$ and $b - b_n > -\varepsilon$, and therefore

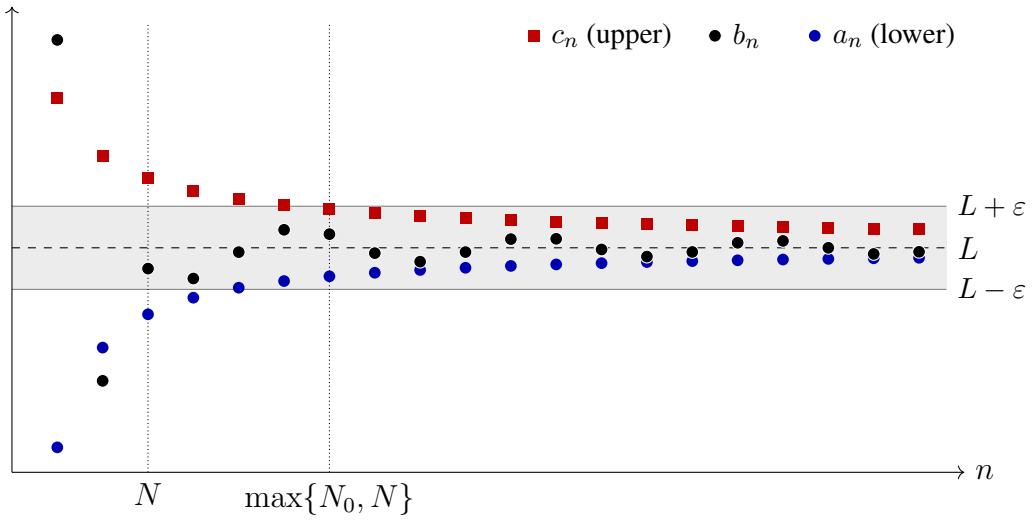
$$a_n - b_n = a_n - a + a - b + b - b_n > -\varepsilon + a - b - \varepsilon = 0$$

for all $n \geq n_0$. So $a_n > b_n$ for $n \geq n_0$. But $n_0 \geq N$, so also $a_n \leq b_n$ for these n, which is a contradiction. □

Remark 2.1. Limits do not preserve strict inequalities. It is **NOT TRUE** that $a_n < b_n$ implies $a < b$. Consider for example $0 < \frac{1}{n}$ for all $n \in \mathbb{N}$, but $\lim 0 = \lim \frac{1}{n} = 0$. What is true is that $a_n < b_n$ implies $a \leq b$.

Theorem 2.2 (Squeeze Theorem). Let (a_n), (b_n), and (c_n) be **real** sequences such that $a_n \to L$ and $c_n \to L$. Suppose $\exists N \in \mathbb{N}$ such that $a_n \leq b_n \leq c_n \ \forall n \geq N$. Then (b_n) converges and $\lim_{n\to\infty} b_n = L$.

The picture to keep in mind: once $n \geq N$, the sequence (b_n) is trapped between (a_n) and (c_n). Eventually both a_n and c_n lie in $(L-\varepsilon, L+\varepsilon)$, so b_n has no choice but to lie there too. (Before N, b_n may do whatever it likes, but finitely many terms do not impact convergence.)



Proof. Let $\varepsilon > 0$. Since $a_n \to L$, there exists $N_1 \in \mathbb{N}$ such that $|a_n - L| < \varepsilon$ for all $n \geq N_1$.
Since $c_n \to L$, there exists $N_2 \in \mathbb{N}$ such that $|c_n - L| < \varepsilon$ for all $n \geq N_2$.
Let $N_0 = \max\{N_1, N_2\}$, so that $|a_n - L| < \varepsilon$ and $|c_n - L| < \varepsilon$ for all $n \geq N_0$. We have

$$\begin{aligned} |a_n - L| < \varepsilon \quad &\iff \quad -\varepsilon < a_n - L < \varepsilon \ \ \forall n \geq N_0 \\ |c_n - L| < \varepsilon \quad &\iff \quad -\varepsilon < c_n - L < \varepsilon \ \ \forall n \geq N_0 \\ a_n \leq b_n \leq c_n \quad &\Rightarrow \ a_n - L \leq b_n - L \leq c_n - L \ \ \forall n \geq N. \end{aligned}$$

Combining these inequalities we obtain for all $n \geq \max\{N_0, N\}$

$$-\varepsilon < a_n - L \leq b_n - L \leq c_n - L < \varepsilon \ \Rightarrow \ |b_n - L| < \varepsilon.$$

We can thus conclude that $b_n \to L$. □

Examples

(i) $\lim \dfrac{\sin n}{n} = 0$, since $-1 \leq \sin n \leq 1 \ \forall n \in \mathbb{N} \Rightarrow -\dfrac{1}{n} \leq \dfrac{\sin n}{n} \leq \dfrac{1}{n}$ and $-\dfrac{1}{n}, \dfrac{1}{n} \to 0$.

(ii) $\lim \dfrac{n}{n!} = 0$, since $\forall n \geq 2$

$$0 \leq \frac{n}{n!} = \frac{1}{(n-1)!} \leq \frac{1}{n-1} \to 0.$$

Proposition 2.1. Let (a_n) and (b_n) be two complex valued sequences. Suppose that (b_n) is bounded and that $a_n \to 0$. Then $a_n b_n \to 0$.

Proof. Since (b_n) is bounded, there exists $M > 0$ such that $|b_n| \leq M$ for all $n \in \mathbb{N}$. Let $\varepsilon > 0$. Since $a_n \to 0$ and $\frac{\varepsilon}{M} > 0$, there exists $n_0 \in \mathbb{N}$ such that

$$|a_n| = |a_n - 0| < \frac{\varepsilon}{M} \quad \forall n \geq n_0.$$

Then, for every $n \geq n_0$,

$$|a_n b_n - 0| = |a_n|\,|b_n| \leq M|a_n| < M \cdot \frac{\varepsilon}{M} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $a_n b_n \to 0$. □

Remark 2.2. The trick is to aim for ε/M instead of ε: the bound M tells us how much $|b_n|$ can magnify $|a_n|$, so we ask $|a_n|$ to be M times smaller in advance. For example, applying Proposition 2.1 directly gives $\frac{\sin n}{n} = \frac{1}{n} \cdot \sin n \to 0$ since $\sin(n)$ is bounded and $\frac{1}{n}$ converges to 0.

3 Algebra of limits

Proposition 3.1. $a_n \to a$ if and only if $a_n - a \to 0$ if and only if $|a_n - a| \to 0$.

Proof. It suffices to write down the three definitions:

(i) $a_n \to a$ if for every $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for every $n \geq n_0$.

(ii) $a_n - a \to 0$ if for every $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that $|(a_n - a) - 0| < \varepsilon$ for every $n \geq n_0$.

(iii) $|a_n - a| \to 0$ if for every $\varepsilon > 0$ there exists $n_0 \in \mathbb{N}$ such that $\big||a_n - a| - 0\big| < \varepsilon$ for every $n \geq n_0$.

Observing that $|a_n - a| = |(a_n - a) - 0| = \big||a_n - a| - 0\big|$ for every $n \in \mathbb{N}$, we see that the three statements say exactly the same thing. □

Proposition 3.2. $a_n \to 0$ if and only if $|a_n| \to 0$.

Proof. A special case of the above Proposition. □

Proposition 3.3. If $a_n \to a$, then $|a_n| \to |a|$.

Proof. Let $\varepsilon > 0$. Since $a_n \to a$, there exists $n_0 \in \mathbb{N}$ such that $|a_n - a| < \varepsilon$ for every $n \geq n_0$. Then, for every $n \geq n_0$,

$$\big||a_n| - |a|\big| \leq |a_n - a| < \varepsilon,$$

by the *reverse triangle inequality* $\big||x| - |y|\big| \leq |x - y|$, which holds for all $x, y \in \mathbb{C}$.
Thus we obtain $|a_n| \to |a|$. □

Remark 3.1. To see that the reverse triangle inequality holds, apply the triangle inequality twice:

$$|x| = |(x - y) + y| \leq |x - y| + |y| \quad \text{and} \quad |y| = |(y - x) + x| \leq |x - y| + |x|.$$

So $|x| - |y| \leq |x - y|$ and $-(|x| - |y|) \leq |x - y|$, which together give us $\big||x| - |y|\big| \leq |x - y|$.

Proposition 3.4. If $a_n \to a$ and $b_n \to b$, then $a_n + b_n \to a + b$.

Proof. Let $\varepsilon > 0$. Since $a_n \to a$, there exists $n_1 \in \mathbb{N}$ such that for every $n \geq n_1$,

$$|a_n - a| < \frac{\varepsilon}{2}.$$

Similarly, since $b_n \to b$, there exists $n_2 \in \mathbb{N}$ such that for every $n \geq n_2$,

$$|b_n - b| < \frac{\varepsilon}{2}.$$

Set $n_0 = \max\{n_1, n_2\}$. Then, for every $n \geq n_0$, we have simultaneously $|a_n - a| < \varepsilon/2$ and $|b_n - b| < \varepsilon/2$. Therefore, for every $n \geq n_0$,

$$\begin{aligned} |(a_n + b_n) - (a + b)| &= |(a_n - a) + (b_n - b)| \\ &\leq |a_n - a| + |b_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, this shows that $a_n + b_n \to a + b$. □

Proposition 3.5. If $a_n \to a$ and $b_n \to b$, then $a_n b_n \to ab$.

Proof. Write

$$a_n b_n - ab = a_n(b_n - b) + b(a_n - a).$$

Observe the following:

(i) (a_n) converges, so it is bounded. Since $b_n - b \to 0$, Proposition 2.1 gives that $a_n(b_n - b) \to 0$.

(ii) Since $a_n - a \to 0$ and the constant sequence (b) is bounded, Proposition 2.1 also gives that $b(a_n - a) \to 0$.

Now Proposition 3.4 shows that

$$a_n(b_n - b) + b(a_n - a) \to 0 + 0 = 0.$$

That is, $a_n b_n \to ab$. □

Proposition 3.6. Let $k \in \mathbb{N}$, $k \geq 2$. If $a_n \to a$, then $a_n^k \to a^k$.

Proof. By induction on k. The base case $k = 1$ is just the hypothesis $a_n \to a$. For the inductive step, suppose that $a_n^m \to a^m$ for some $m \geq 1$. Then

$$a_n^{m+1} = a_n \cdot a_n^m \to a \cdot a^m = a^{m+1},$$

by Proposition 3.5. Hence $a_n^k \to a^k$ for every $k \in \mathbb{N}$, and in particular for every $k \geq 2$. □

Proposition 3.7. Let (a_n) and (b_n) be sequences with $b_n \neq 0$ for every $n \in \mathbb{N}$. If $a_n \to a$ and $b_n \to b \neq 0$, then $a_n/b_n \to a/b$.

Proposition 3.8. Let $k \in \mathbb{N}$, $k \geq 2$. If $a_n \geq 0$ for every $n \in \mathbb{N}$ and $a_n \to a$, then $\sqrt[k]{a_n} \to \sqrt[k]{a}$.

The proofs of those propositions can be found in the supplemental lecture notes.

Example 3.1. Using the Propositions above, find the following limits.

(i) $\lim_{n\to\infty}\left(5+\frac{1}{n}\right)$.

(ii) $\lim_{n\to\infty}\frac{1}{n^2}$.

(iii) $\lim_{n\to\infty}\frac{5}{n^2}$.

(iv) $\lim_{n\to\infty}\frac{n^3+2n}{5n^3+2}$.

4 Divergence to infinity

Definition 4.1. We say that a real sequence (a_n) **diverges to** ∞ if

$$\forall M>0 \;\; \exists N(M)\in\mathbb{N} \;\; \text{such that} \;\; a_n \geq M \;\; \forall n \geq N(M).$$

We write $\lim a_n = \infty$. Similarly, (a_n) **diverges to** $-\infty$ if

$$\forall M>0 \;\; \exists N(M)\in\mathbb{N} \;\; \text{such that} \;\; a_n \leq -M \;\; \forall n \geq N(M).$$

Theorem 4.1. Let $(a_n),(b_n)$ be real sequences and suppose $\exists N\in\mathbb{N}$ such that $a_n \leq b_n \; \forall n \geq N$. Then

(i) If $\lim a_n = \infty$ then $\lim b_n = \infty$

(ii) If $\lim b_n = -\infty$ then $\lim a_n = -\infty$.

Proof.

□

Example 4.1. $\lim \frac{n^2-3}{n} = \infty$ since

$$\frac{n^2-3}{n} = n - \frac{3}{n} \geq n-1 \; \forall n \geq 3,$$

and $\lim(n-1)=\infty$ (**why?**).